

**Deliberation and Public Spiritedness:
Consulting the public in Texas on Utility Choices**

Rui Wang

Stanford University

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Abstract

One of the hypotheses about the nature of post-deliberation preferences is that they will be more “public spirited.” But whether or not it functions that way is an empirical question. Yet, there is little evidence to establish the link between deliberation and public spiritedness. In seven deliberative polls on electricity issues, participants were asked how much more, if anything, they would be willing to pay on their monthly electric bill for renewable energy and energy conservation. In three of the seven polls, they were also asked how important it is to have everyone’s basic needs met, even if that means the rest of them have to pay a little more. Using these variables as proxy measures of public spiritedness, this paper aims to explain how, if any, deliberation promotes public spiritedness.

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The long debate about whether ordinary citizens are capable of governing themselves has not yet achieved an agreement among political theorists. Some believe that the masses are incapable of rational arguments and they cannot see beyond their narrow selfish concerns (see Mendelberg, 2002; Sanders, 1997). Others insist that deliberation is expected to produce a variety of positive democratic outcomes, e.g., a more empathic view of the other, a better-informed perspective on public problems, and a broader understanding of one's interest (see Mendelberg, 2002). In this way, deliberative democracy can serve the common good.

J. S. Mill advocates for more “school of public spirit,” where citizens discuss public issues together and take responsibilities for the broader public interest. He argues that when ordinary citizens participate in public functions, “He is called upon, while so engaged, to weigh interests not his own; to be guided in case of conflicting claims, by another rule than his private partialities; to apply, at every turn, principles and maxims which have for their reason of existence the common good He is made to feel himself one of the public, and whatever is their interest to be his interest” (p. 79, J. S. Mill, 1861).

One of the hypotheses about the nature of post-deliberation preferences is that people will be more “public spirited.” Mendelberg (2002) once pointed out: “the promise of deliberation is its ability to foster the egalitarian, reciprocal, reasonable and open-minded exchange of language” (p.153). Fishkin (2009) argues that when people share their reasons in a dialogue about public problems, everyone is sensitized to broader public concerns. They come to

understand the interest and value at state from the perspective of other members of the community.

Mill called for more “schools of public spirit” and experimentation with their design. Theoretically in a sense the Deliberative Polling (DP), like other microcosmic deliberations, is potentially a “school of public spirit” (Fishkin, 2009). But whether or not it functions that way is an empirical question. Yet, there is little evidence to establish the link between deliberation and public spiritedness. Does deliberation promote public spiritedness?

Seven deliberative polls on utility choices were conducted in Texas between 1996 and 1998. In each poll, participants were asked how much more, if anything, they would be willing to pay on their monthly electric bill for renewable energy and energy conservation. In three of the seven polls, they were also asked how important it is to have everyone’s basic needs met, even if that means the rest of them have to pay a little more. Using these variables as proxy measures of public spiritedness, this paper aims to explain how, if any, deliberation promotes public spiritedness.

Data

In order to decide the best way to meet the increasing demand for electricity, eight energy companies, including CPL, WTU, SWEPCO, HLP, ENTERGY, TU, SPS & EPE, invited their customers from Texas and adjacent regions to deliberate on utility choices between 1996 and 1998. Over 10,000 randomly selected customers participated in the initial survey. About 200 or so customers took part in the two-day deliberation event in each DP and completed the post deliberation survey.

Almost identical questions were asked in CPL, WTU and SWEPCO DPs. These three datasets were merged together to serve as the main dataset for our primary analysis. Since some of the data from EPE were missing due to an accident, this DP was excluded from our analysis. The other four DPs (HLP, ENTERGY, TU, and SPS) were analyzed secondarily to supplement the primary analysis. Table 1 below presents a brief summary of the respondent and non-respondent samples of our primary and secondary datasets.

[Table 1 about here]

Table 2 shows the representativeness of our samples based on comparison of the demographic distributions of the respondent and non-respondent samples. Overall, DP participants are not significantly different from non-participants, except that lower-educated and older people are slightly under-represented.

[Table 2 about here]

Do people become more public-spirited after deliberation?

During the two-day deliberation events, participants discussed the benefits and disadvantages of different energy options, including offering energy efficiency programs, building new fossil fuel plants, building new renewable energy plants, purchasing power from outside sources, etc. In the survey, participants were asked how much more, if any, they would be willing to pay on their monthly electric bill for their utility company to include each option in its plan to meet the area's need for electricity. Table 3 shows the percentage of DP participants

who would be willing to pay at least one dollar more on their monthly electric bill to implement any of the energy options before and after deliberation, as well as the increase in the percentage. Overall, more than 1/3 of the DP participants in our primary data would be unwilling to pay anything more on their electric bill for any of the energy options before deliberation (more than half in our secondary data). These numbers reduced to less than 10% after deliberation in our primary data, and 20% in our secondary data. In other words, more than one quarter of the participants in our primary data who initially refused to pay additional dollars for any energy options became more generous after discussing and thinking through electricity needs and options (30% for our secondary data).

[Table 3 about here]

Indirect measurement of public spiritedness

Even though DP participants became more generous after deliberation for energy options in general, the percentage of people who were willing to contribute at least one dollar more on their monthly electric bill increased significantly for two utility options: one is building new renewable energy plants, and the other is energy conservation.

In each of the DPs, participants were asked how much more they were willing to pay for renewable energy and energy conservation (the exact wording of these questions might be slightly different in different DPs). For example, in CPL, participants were asked:

One effect of CPL incorporating some of the options mentioned above could be an electric bill for customers that is slightly higher in the long run. Now we want to ask you how much more, if anything, you would be willing to pay on your monthly electric bill for

your utility to meet each goal. If you are unwilling to pay any more just write 0. Please answer in terms of dollars per monthly bill.

- *Generation by technologies such as wind and solar power.*
- *Services and technologies which reduce the need for additional electric generation facilities.*

As Fishkin (2009) noted, the notion that one is willing to pay more to subsidize the cost of renewable energy and conservation programs in some sense indicate that one is willing to contribute to the broader public interest. Therefore, these two variables are considered as indirect measures of public spiritedness.

Table 4A and 4B show the increase in the percentage of people who were willing to pay at least one dollar more for renewable energy after deliberation. In our primary data, as can be seen in Table 4A, the overall percentage of people who were willing to pay at least one dollar more for wind and solar power increased significantly from 56.1% to 84.5% after deliberation. The increases in the percentage of people who were willing to pay at least one dollar more for renewable energy range from 22.0% to 32.6% for CPL, WTU, and SWEPCO, with an average of 28.5%. All the increases after deliberation are statistically significant ($p < .001$). Our secondary data (4B) show similar increases. The average increase across HLP, ENTERGY, TU and SPS is 25.4% ($p < .001$).

[Table 4 about here]

Table 4C and 4D show the increase in the percentage of people who were willing to pay at least one dollar more for energy conservation. Similarly, the percentage increased

significantly after deliberation. The overall increase in our primary data (4C) is 27.9%, ranging from 20.7% to 34.0%, and 22.9% in our secondary data (4D), ranging from 20.1% to 30.1%. All the increases are statistically significant at .001 level.

In the DPs, participants were also asked how much more they were willing to pay for coal or natural gas.

- *Generation using fuels like natural gas or coal. (CPL, WTU & SWEPCO)*
- *Electric generation from facilities that use coal or natural gas. (HLP & ENTERGY)*
- *Electric generation from facilities that use coal. (TU & SPS)*
- *Electric generation from facilities that use natural gas. (TU & SPS)*

The percentage of people who were willing to pay at least one dollar more did not increase for all the utility options. Table 4E-4H present the results for coal and natural gas. When coal was mentioned alone (as in TU & SPS) or first (as in HLP & ENTERGY), people were not significantly more willing to pay one dollar more for this option after deliberation. When natural gas was mentioned alone (as in TU & SPS) or first (as in CPL, WTU & SWEPCO), the percentage of people who were willing to pay at least one dollar more increased significantly (except for TU, where no change was observed). However, the magnitude of the increase is less than half of that in renewable energy or conservation.

Clear-cut indicator of public spiritedness

In three polls (CPL, WTU, and SWEPCO only), DP participants were asked the following question: *Many things that some people think are important may have some cost or tradeoff. For each of the following, please rate how important you think it is, given the tradeoff,*

using a 0 to 10 scale where 0 stands for not at all important, 10 stands for extremely important, and 5 stands for average importance.

- *To see to it everyone has their basic needs met, even if that means the rest of us have to pay a little more.*

Based on our definition, this question is a clear-cut measure of public spiritedness. Table 5 shows DP participants' rating of the importance of the above tradeoff question before and after deliberation, as well as the increase. For all three DPs, after deliberation, participants thought it was more important to see to it everyone has their basic needs met, even if that means the rest of them have to pay a little more. All the increases in the rating of importance are statistically significant ($p \leq .001$).

[Table 5 about here]

How does deliberation promote public spiritedness?

We have found evidence that people became more public-spirited after deliberation. Our next question is “how does deliberation promote public spiritedness?”

Method

Explanatory variables

Values:

- 1) Environmental protection index (available in our primary data only): this index was computed as the mean importance rating of the following three items: 1) to reduce pollution; 2) using resources that maintain current levels of environmental quality; and

(3) options that add an extra measure of environmental protection. DP participants were asked to rate how important they think it is for each item, using a 0 to 10 scale, where 0 stands for not at all important, 10 stands for extremely important, and 5 stands for average importance. Factor analysis was performed and found these three items were highly correlated (Cronbach's alpha is .684 at T1 and .675 at T2). The environmental protection index was rescaled into a 10-point scale on 0-1, where 1 indicates valuing the environment as extremely important, 0 indicates valuing the environment as not at all important, and 0.5 stands for average importance.

- 2) Single environmental protection item (available in our secondary data only): DP participants were asked to rate how important they think it is to protect the environment from pollution created by electric generation. This item was rescaled into a 10-point scale on 0-1, where 1 indicates extremely important, 0 indicates not at all important, and 0.5 stands for average importance.
- 3) Conservation index (available in our primary data only): this index was calculated as the mean of the respondent's importance rating of the following three items: 1) to preserve resources like coal, natural gas, and oil for the future, 2) for utilities to reduce their use of natural gas and coal by helping their customers use energy more efficiently, and 3) services and technologies which reduce the need for additional electric generation facilities. Factor analysis was performed and found these three items were highly correlated (Cronbach's alpha is .523 at T1 and .547 at T2). The conservation index was rescaled into a 10-point scale on 0-1, where 1 indicates valuing conservation as extremely important, 0 indicates valuing conservation as not at all important, and 0.5 stands for average importance.

- 4) Economic growth (available in our primary data only): in the survey, DP participants were also asked to rate the importance of promoting economic growth in the community. This item was rescaled into a 10-point scale on 0-1, where 1 indicates extremely important, 0 indicates not at all important, and 0.5 stands for average importance.
- 5) Lowest cost (available in both primary and secondary data): DP participants were also asked to rate the importance of using resources which result in the lowest cost electricity. This item was also rescaled into a 10-point scale on 0-1, where 1 indicates extremely important, 0 indicates not at all important, and 0.5 stands for average importance.
- 6) Ideology (available in our primary data only): this continuous variable was rescaled into a 10-point scale on 0-1, where 1 indicates liberal, 0 indicates conservative, and 0.5 stands for exactly in the middle.

Empirical premises (available in our primary data only):

In the survey, DP participants were asked *if they agree strongly, agree mildly, neither agree nor disagree, disagree mildly, or disagree strongly with the following statements:*

- *Using fuels like coal and natural gas has a negative impact on the environment.*
- *CPL (WTU, SWEPCO) is doing enough to develop renewable energy options like wind and solar power and to develop ways to use energy more effectively.*
- *Higher electric bills hinder economic development.*
- *Higher electric bills lead people to use less electricity.*
- *Increasing use of solar and wind power will make these options less expensive in the future.*

Each item of the empirical premises was rescaled into a 5-point scale on 0-1, where 1 indicates agree strongly, 0 indicates disagree strongly, and 0.5 stands for neither agree nor disagree.

Knowledge index (available in our primary data only):

In the survey, DP participants were asked a series of factual questions, to gauge their general knowledge about the topic being discussed: *The following questions are about the production and use of electricity. We realize that not everybody has thought much about these things, so please just tell us if you don't know.*

- *What energy source do you think produces the greatest amount of CPL's (WTU's, SWEPCO's) electricity; coal, wind, natural gas, fuel oil, nuclear, or solar?*
- *Overall, what group of CPL (WTU, SWEPCO) customers consume the most kilowatt hours of electricity: residential, commercial, or industrial?*
- *Which customer group pays the highest electric rates; residential, commercial, or industrial?*
- *Which of the following resources used to produce electricity produces the most air emissions; nuclear, coal, or natural gas?*
- *Which agency sets electric rates in Texas?*

For each knowledge item, DP participants were given a score of either 1, for giving a correct answer, or 0, for an incorrect answer or not giving an answer at all. The knowledge indices (at T1 and T2) were calculated as the mean score the respondent got for the above knowledge questions asked in the survey. The final scores range from 0 to 1, indicating the percentage of knowledge items that the participants answered correctly. Two CPL-only

knowledge items were excluded from constructing the knowledge index for the purpose of comparison and consistency.

The average knowledge score of the DP participants is 0.4 before deliberation, meaning that on average DP participants answered only 2 out of the 5 knowledge questions correctly. After deliberation, the average knowledge score increased to 0.64, meaning that DP participants gave correct answers to more than 3 knowledge questions on average. The knowledge gain after deliberation is significant at .001 level.

Analysis

Zero-inflated Poisson regression. Two indirect measures of public spiritedness – the amount of dollars people are willing to pay more for renewable energy and conservation – are count variables, containing non-negative integers with many zeros. A zero-inflated Poisson model is appropriate to estimate such count variables. Zero-inflated Poisson model has two components – one is to estimate the actual count by fitting a Poisson regression model and the other is to estimate the excess of zeros by fitting a logistic regression model (Lambert, 1992).

$$Pr(y_i=0)=\pi_i+(1-\pi_i)e^{-\lambda_i}$$

$$Pr(y_i=k_i)=(1-\pi_i)\frac{\lambda_i^{k_i}e^{-\lambda_i}}{k_i!}, \quad k_i=1, 2, 3, \dots, \text{ where the outcome}$$

variable y_i represents how much more people are willing to pay for renewable energy or conservation; λ_i is the expected Poisson count for the i^{th} individual; π_i is the probability of extra zeros.

Results

Public spiritedness

Table 6 shows the relationship between the clear-cut indicator of public spiritedness and the amount of dollars DP participants were willing to pay more on their monthly electric bill for renewable energy and conservation before and after deliberation. The zero-inflation model shows the coefficients of the logistic regression estimating the excess of zeros. The count model shows the coefficients of the Poisson regression estimating the positive amount of dollars DP participants were willing to pay more on their monthly electric bill (excluding all the zeros). Before deliberation, DP participants who were more public-spirited were significantly more willing to pay at least one dollar more for renewable energy (odds ratio=2.3, $p=.002$). After deliberation, the odds ratio for DP participants to pay at least one dollar more increased to 2.9 ($p=.006$). At the meantime, DP participants were willing to pay more dollars on their monthly electric bill for renewable energy after deliberation as they became more public-spirited (as one unit increase in public spiritedness, the amount of dollars DP participants were willing to pay more for renewable energy increased by 1.64 times after deliberation, $p<.001$).

[Table 6 about here]

Similarly, the amount of dollars DP participants were willing to pay more for energy conservation was also positively associated with the clear-cut indicator of public spiritedness. As Table 6B shows, more public-spirited DP participants were willing to pay more for conservation both before and after deliberation. Meanwhile, the amount of dollars DP participants were willing to pay more on their monthly electric bill for conservation was more positively associated with public spiritedness after deliberation.

Renewable energy

Our next step is to examine the relationship between the attitudes and values of DP participants and the amount of dollars they were willing to pay more on their monthly electric bill for renewable energy and conservation. Table 7A presents the result of the zero-inflated Poisson regression of the amount of dollars DP participants were willing to pay more for renewable energy on their values. As expected, environmental protection index and lowest cost are significant predictors of how much more DP participants were willing to pay for renewable energy. Both before and after deliberation, those who were more concerned about the environment were more willing to pay at least one dollar more for renewable energy compared to those who were less concerned about the environment (the odds ratio increased from 6.86 before deliberation to 28.02 after deliberation). At the same time, those who were more concerned about the cost of electricity were less likely to pay more for renewable energy (the odds ratio decreased from 0.21 to 0.01). The positive association between the concerns about the environment and the amount of dollars people were willing to pay more for renewable energy became stronger after deliberation, so did the negative association between the concerns of the cost and the amount they were willing to pay more. The count model suggested similar trend, where the actual amount of dollars DP participants were willing to pay more for renewable energy was positively associated with environment concerns and negatively associated with cost concerns.

Table 7B shows the relationship between DP participants' values and the amount of dollars they were willing to pay more for renewable energy in our secondary data. The results are consistent with the evidence presented earlier for our primary data. Both before and after

deliberation, cost concern is negatively associated with the amount of dollars they were willing to pay more, while the concern of protecting the environment from pollution created by electric generation is positively associated with willingness to pay more. However, the associations became much stronger after deliberation. For example, in the zero-inflation model estimating the likelihood of paying nothing, the odds ratio increased from 4.17 to 19.6 as one unit increase in the rating of the importance of using resources which result in the lowest cost electricity. Similarly, DP participants were much more willing to pay at least one dollar more on their electric bill for renewable energy after deliberation compared to before deliberation as they became more concerned about the environment (the odds ratio increased from 2.9 to 16.1 with one unit increase in the rating of importance of protecting the environment from pollution created by electric generation).

Similar patterns were observed in the count model estimating the actual amount of dollars DP participants were willing to pay more for conservation. As can be seen in Table 7B, the positive association between their rating of the importance of protecting the environment and the amount of dollars they were willing to pay more for conservation became much stronger after deliberation (as one unit increase in the rating of environment protection, the amount of dollars DP participants were willing to pay more increased by 1.55 times before deliberation, and 2.83 times after deliberation). Similarly, the magnitude of the negative association between the amount of dollars DP participants were willing to pay more for conservation and their rating of the importance of using resources which result in the lowest cost electricity increased after deliberation (as one unit increase in the importance rating of cost, the amount of dollars DP participants were willing to pay more decreased to 88%, $p=.086$ before deliberation, and 27%, $p<.001$ after deliberation).

[Table 7 about here]

Table 8A shows the relationship between empirical premises and the amount of dollars DP participants were willing to pay more for renewable energy. As expected, the amount of dollars people are willing to pay more for renewable energy is associated with their agreement or disagreement with the following statements: 1) using fuels like coal and natural gas has negative impact on environment; 2) CPL (WTU, SWEPCO) is doing enough for renewable energy and conservation; and 3) increasing use of wind or solar power will make it less expensive in future. After deliberation, participants who agreed that increasing the use of renewable energy will make it less expensive in the future became more willing to pay at least one dollar more for renewable energy (the odds ratio increased from 1.93 before deliberation to 8.30 after deliberation). Those who thought using fuels has negative impact on environment became more willing to pay more for renewable energy (odds ratio increased from 1.76 to 3.8). Meanwhile, those who believed that CPL (WTU, SWEPCO) is doing enough for renewable energy and conservation became less willing to pay more for renewable energy after deliberation (odds ratio=0.36, $p=.005$).

[Table 8 about here]

Table 9A shows the relationship between knowledge and the amount of dollars DP participants were willing to pay more for renewable energy. As can be seen, DP participants became more willing to pay at least one dollar more for renewable energy as they gained more knowledge about different utility choices after deliberation (odds ratio increased from 3.01

before deliberation to 6.01 after). More importantly, they were more willing to pay more dollars for renewable energy as they learned more after deliberation (as one unit gain in knowledge, the amount of dollars DP participants were willing to pay more for renewable energy increased by 1.23 times after deliberation, $p < .001$, as compared to the negative association before deliberation).

[Table 9 about here]

Conservation

Our second proxy measure of public spiritedness is the amount of dollars DP participants were willing to pay more on their monthly electric bill for energy conservation programs. Table 7C shows the relationship between DP participants' values and how much more they were willing to pay for conservation. As expected, DP participants who valued more about energy efficiency were not only more likely to pay at least one dollar more for conservation after deliberation, they were also more likely to pay more dollars (the amount of dollars DP participants were willing to pay more increased by 2.14 times, $p < .001$, after deliberation, as one unit increase in the importance rating of energy efficiency). The likelihood of paying nothing for conservation for those who were more concerned about the cost of electricity became greater after deliberation (the odds ratio increased from 2.9 to 4.7 with one unit increase in the rating of the importance of using resources which result in the lowest cost electricity).

Table 7D shows the relationship between DP participants' values and the amount of dollars they were willing to pay more for conservation in our secondary data. Similar to what we found for renewable energy, the amount of dollars DP participants were willing to pay more for

conservation was also negatively associated with their importance rating of using resources which result in the lowest cost electricity and positively related to their ratings of protecting the environment from pollution created by electric generation. But after deliberation, both associations became much stronger. For example, in the zero-inflation model estimating the likelihood of paying nothing, the odds ratio increased from 6.82 to 16.79 as one unit increase in the rating of the importance of using resources which result in the lowest cost electricity, and decreased from 0.26 to 0.13 as one unit increase in the rating of importance of protecting the environment from pollution after deliberation. Also, DP participants who were more concerned about the environment were willing to spend more money on conservation after deliberation (as one unit increase in the importance rating of protecting the environment, the amount of dollars DP participants were willing to pay more increased by 1.18 times, $p=.039$ before deliberation, and 1.58 times, $p<.001$ after deliberation).

Table 8B shows the relationship between empirical premises and the amount of dollars DP participants were willing to pay more for energy conservation programs. As expected, the amount of dollars people were willing to pay for energy conservation is associated with their agreement or disagreement with the following statements: using fuels has negative impact on environment. After deliberation, DP participants who believed that using fuels has negative impact on environment became more willing to pay at least one dollar more for conservation (odds ratio increased from 1.91, $p=.018$ before deliberation to 2.55, $p<.001$ after deliberation). Meanwhile, people who agreed that increasing use of renewable energy will make it less expensive in the future were not only more willing to pay at least one dollar more for conservation after deliberation (odds ratio=2.05, $p=.026$), they were also willing to pay more

dollars (the amount of dollars they were willing to pay more increased by 1.92 times, $p < .001$ as one unit of increase in the agreement).

Table 9B shows the relationship between knowledge and the amount of dollars DP participants were willing to pay more for energy conservation. Even though the magnitude of the significantly positive association between knowledge and the willingness of paying at least one dollar more for conservation did not increase after deliberation, DP participants became more willing to pay greater amount of dollars for conservation as they learned more after deliberation (as one unit gain in knowledge, the amount of dollars DP participants were willing to pay more for conservation increased by 1.33 times after deliberation, $p < .001$, as compared to the significantly negative association before deliberation).

Discussion

Based on our findings from seven Texas DPs, deliberation did show positive impacts on the general public. First of all, DP participants who initially refused to contribute additional dollars to help utility companies meet increasing electricity needs became more generous after deliberation. Before deliberation, people may not be aware of the increasing demand for electricity and the urgency of implementing new energy options to respond to this demand. After deliberating on electricity issues as well as reading briefing documents presenting pros and cons of different energy alternatives, they formed a more considered opinions on the current status of the electricity demand and supply, and realized the necessity for utility companies to incorporate new energy options to meet their electricity needs. As a result, DP participants became more willingly to contribute money to sponsor new energy plans. In other words, people

became more public-spirited after deliberation about meeting the electricity needs of the whole community, even if that means they have to pay more.

Secondly, among all the potential energy options, building renewable energy plants and offering energy efficiency programs are the two options that won most support from DP participants. Compared to other energy options, such as building new fossil fuel plants such as coal or natural gas or buying power from outside suppliers, switching to renewable energy plants or energy conservation programs obviously have significant long-term benefits to both the human-being and the environment.

Thirdly, people who wanted to pay more after deliberation were not making random decisions. Our findings suggest that they were willing to pay more because their attitudes and values changed after deliberation. Post deliberation, the predicting power of the variables positively or negatively related to the amount of dollars DP participants were willing to pay more for renewable energy or conservation became much stronger, suggesting that DP participants became more rational in making their decisions after deliberation.

Finally, the willingness of paying more dollars for renewable energy and conservation is significantly positively associated with participant's knowledge after deliberation. In other words, the more they learned from deliberation, the more public-spirited they became.

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Table 1
Participant and Non-participant Samples

A. CPL, WTU, SWEPCO

DP	DP participants	Non-participants
CPL	216 (17.3%)	1,030 (82.7%)
WTU	230 (18.7%)	1,000 (81.3%)
SWEPCO	232 (15.7%)	1,246 (84.3%)
Total	678 (17.1%)	3,276 (82.9%)

B. HLP, ENTERGY, TU, SPS

DP	DP participants	Non-participants
HLP	192 (22.7%)	655 (77.3%)
ENTERGY	175 (17.5%)	825 (82.5%)
TU	226 (17.4%)	1,072 (82.6%)
SPS	222 (16.7%)	1,108 (83.3%)
Total	815 (18.2%)	3,660 (81.8%)

A.

Table 2
Representativeness

A. CPL, WTU, SWEPCO

		DP participant s	Non- participants	<i>p</i>
Gender	Male	44.6%	41.5%	.130
Education	0-8th	1.6%	4.3%	<.001
	9-11th	7.0	8.2	
	H.S. Graduate	28.4	34.0	
	Some College	33.0	26.7	
	College Graduate	19.9	18.4	
	Post-Grad Work	10.1	8.3	
Income	Less than \$6000	6.3%	6.2%	.468
	\$6000 - 10000	7.1	8.1	
	\$10001 - 15000	11.1	11.2	
	\$15001 - 25000	20.1	18.3	
	\$25001 - 35000	18.9	19.2	
	\$35001 - 55000	22.9	20.2	
	\$55001 - 75000	8.5	10.1	
	More than \$75000	5.1	6.8	
Race	Hispanic	16.6%	19.0%	.081
	Black	9.1	6.7	
	Asian	0.7	0.6	
	Anglo	68.8	70.1	
	Other	4.6	3.6	
Age (Mean)		45.7	47.8	.002

Note: All *p*'s are from two-tailed tests.

B. HLP, ENTERGY, TU, SPS

		DP participants	Non- participants	<i>p</i>
Gender	Male	44.8%	43.7%	.568
Education	Less than High School	2.7%	3.6%	<.001
	Some High School	6.5	7.8	
	H.S. Graduate	20.2	29.6	
	Some College	28.3	25.1	
	College Graduate	27.2	22.1	
	Graduate School	11.4	8.7	
	Trade/Technical School	3.7	2.9	
Income	Less than \$15K	14.2%	13.3%	.107
	\$15K - \$25K	16.5	18.6	
	\$25K - \$35K	17.3	17.0	
	\$35K - \$50K	24.6	20.4	
	\$50K - \$75K	15.8	16.7	
	\$75K - \$100K	6.7	7.3	
	More than \$100K	5.0	6.7	
Race	Hispanic	13.2%	13.4%	.514
	Black	10.7	9.0	
	Asian	1.4	1.2	
	Anglo	68.4	70.5	
	Native American	4.0	3.2	
	Other	2.4	2.8	
Age	18-25	6.3%	8.7%	.001
	26-35	19.1	18.1	
	36-45	27.1	24.9	
	46-55	20.2	19.4	
	56-65	15.3	12.0	
	Over 65	12.1	16.9	

Note: All *p*'s are from two-tailed tests.

Table 3
Percentage Willing to Pay at Least \$1 More per Month to Meet the Area's Electricity Need

A. CPL, WTU, SWEPCO

	T1	T2	T2-T1	<i>p</i>	N (missing)
CPL	70.1%	88.7%	18.6%	<.001	194 (22)
WTU	66.8	92.4	25.6	<.001	223 (7)
SWEPCO	59.5	92.8	33.3	<.001	222 (10)
Overall	65.3	91.4	26.1	<.001	639 (39)

B. HLP, ENTERGY, TU, SPS

	T1	T2	T2-T1	<i>p</i>	N (missing)
HLP	52.1%	84.2%	32.2%	<.001	171 (21)
ENTERGY	43.8	81.7	37.9	<.001	169 (6)
TU	55.4	78.9	23.5	<.001	213 (13)
SPS	48.8	76.1	27.3	<.001	205 (17)
Overall	50.3	80.0	29.7	<.001	758 (57)

Note: All *p*'s are from one-tailed tests.

Table 4
Percentage Willing to Pay at Least \$1 for Specific Options

A. Renewable Energy (CPL, WTU, SWEPCO)

	T1	T2	T2-T1	<i>p</i>	N (missing)
CPL	59.9%	81.9%	22.0%	<.001	182 (34)
WTU	55.4	87.9	32.6	<.001	215 (15)
SWEPCO	53.6	83.4	29.9	<.001	211 (21)
Overall	56.1	84.5	28.5	<.001	608 (70)

B. Renewable Energy (HLP, ENTERGY, TU, SPS)

	T1	T2	T2-T1	<i>p</i>	N (missing)
HLP	43.4%	69.7%	26.3%	<.001	175 (17)
ENTERGY	37.0	66.7	29.7	<.001	165 (10)
TU	62.9	87.3	24.4	<.001	213 (13)
SPS	58.4	80.7	22.3	<.001	202 (20)
Overall	51.5	77.0	25.4	<.001	755 (60)

C. Conservation (CPL, WTU, SWEPCO)

	T1	T2	T2-T1	<i>p</i>	N (missing)
CPL	51.7%	72.4%	20.7%	<.001	174 (42)
WTU	42.7	70.6	28.0	<.001	211 (19)
SWEPCO	45.9	79.9	34.0	<.001	209 (23)
Overall	46.5	74.4	27.9	<.001	594 (84)

D. Conservation (HLP, ENTERGY, TU, SPS)

	T1	T2	T2-T1	<i>p</i>	N (missing)
HLP	44.6%	64.9%	20.2%	<.001	168 (24)
ENTERGY	32.5	62.6	30.1	<.001	163 (12)
TU	57.4	77.5	20.1	<.001	209 (17)
SPS	49.8	71.9	22.1	<.001	199 (23)
Overall	47.0	69.8	22.9	<.001	739 (76)

Note: All *p*'s are from one-tailed tests.

E. Natural Gas or Coal (CPL, WTU, SWEPCO)

	T1	T2	T2-T1	<i>p</i>	N (missing)
CPL	38.2%	49.4%	11.2%	.034	170 (46)
WTU	34.8	48.4	13.6	.001	221 (9)
SWEPCO	28.8	44.8	16.0	<.001	212 (20)
Overall	33.7	47.4	13.8	<.001	603 (75)

Note: All *p*'s are from two-tailed tests.

F. Coal or Natural Gas (HLP, ENTERGY)

	T1	T2	T2-T1	<i>p</i>	N (missing)
HLP	25.4%	27.8%	2.4%	.539	169 (23)
ENTERGY	24.7	27.8	3.1	.477	162 (13)
Overall	25.1	27.8	2.7	.346	331 (36)

Note: All *p*'s are from two-tailed tests.

G. Coal (TU, SPS)

	T1	T2	T2-T1	<i>p</i>	N (missing)
TU	11.7%	15.0%	3.3%	.308	214 (12)
SPS	23.5	27.5	4.0	.250	200 (22)
Overall	17.4	21.0	3.6	.128	414 (34)

Note: All *p*'s are from two-tailed tests.

H. Natural Gas (TU, SPS)

	T1	T2	T2-T1	<i>p</i>	N (missing)
TU	42.5%	42.5%	0%	1.000	212 (14)
SPS	25.6	38.5	12.8	.002	195 (27)
Overall	34.4	40.5	6.1	.038	407 (41)

Note: All *p*'s are from two-tailed tests.

Table 5
Public spiritedness

CPL, WTU, SWEPCO

	T1	T2	T2-T1	<i>p</i>	N (missing)
CPL	6.33	6.99	.66	<.001	205 (11)
WTU	6.77	7.63	.86	<.001	226 (4)
SWEPCO	6.78	7.51	.74	<.001	227 (5)
Overall	6.64	7.39	.75	<.001	658 (20)

Note: All *p*'s are from one-tailed tests.

Public spiritedness is measured by the following item: *please rate how important you think it is, given the tradeoff, using a 0 to 10 scale where 0 stands for not at all important, 10 stands for extremely important, and 5 stands for average importance. To see to it everyone has their basic needs met, even if that means the rest of us have to pay a little more.*

Table 6
Willingness to Pay More for Specific Options as a Function of Public Spiritedness

A. Renewable Energy (CPL, WTU, SWEPCO)

Predictor	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Public spiritedness	.098	.064	-.831	.002	.496	<.001	-1.034	.006
Likelihood ratio	-2073				-3152			

B. Conservation (CPL, WTU, SWEPCO)

Predictor	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Public spiritedness	.229	<.001	-.857	.001	.540	<.001	-.635	.040
Likelihood ratio	-2056				-2239			

Note: All *p*'s are from one-tailed tests.

Table 7
Willingness to Pay More for Specific Options as a Function of Values

A. Renewable Energy (CPL, WTU, SWEPCO)

Value	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Environmental protection index	.723	<.001	-1.926	.002	.932	<.001	-3.333	<.001
Lowest cost	-.525	<.001	1.544	.003	-.369	<.001	4.311	<.001
Promote economic growth	-.385	<.001	-.073	.872	.079	.357	.952	.223
Ideology	-.027	.684	.285	.361	-.182	.002	-.356	.419
Likelihood ratio	-1931				-2813			

B. Renewable Energy (HLP, ENTERGY, TU, SPS)

Value	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Environmental protection	.439	<.001	-1.067	.002	1.039	<.001	-2.777	<.001
Lowest cost	-.131	.086	1.429	.004	-1.322	<.001	2.978	<.001
Likelihood ratio	-3615				-4449			

C. Conservation (CPL, WTU, SWEPCO)

Value	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Conservation index	-.180	.274	-3.632	<.001	.759	<.001	-3.505	<.001
Lowest cost	.467	<.001	1.533	.003	.086	.304	1.676	<.001
Promote economic growth	-.842	<.001	.690	.135	-.026	.812	.287	.617
Ideology	-.332	<.001	.369	.236	.024	.751	.277	.440
Likelihood ratio	-1940				-2033			

D. Conservation (HLP, ENTERGY, TU, SPS)

Value	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Environmental protection	.168	.039	-1.348	<.001	.455	<.001	-2.048	<.001
Lowest cost	-.078	.226	1.920	<.001	.122	.092	2.821	<.001
Likelihood ratio	-3311				-3657			

Note: All *p*'s are from one-tailed tests.

Table 8
Willingness to Pay More for Specific Options as a Function of Empirical Premises

A. Renewable Energy (CPL, WTU, SWEPCO)

Empirical premise	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Using fuels has negative impact on environment	.109	.047	-.566	.035	.063	.115	-1.338	<.001
CPL (WTU, SWEPCO) is doing enough for renewable energy and conservation	.115	.035	.280	.185	.065	.098	1.030	.005
Higher electric bills hinder economic development	-.498	<.001	1.032	.001	-.349	<.001	.381	.194
Higher electric bills lead people to use less electricity	-.018	.388	.217	.248	-.189	<.001	.821	.045
Increasing use of renewable will make it less expensive in future	.363	<.001	-.659	.040	.258	<.001	-2.116	<.001
Likelihood ratio	-1095				-2779			

B. Conservation (CPL, WTU, SWEPCO)

Empirical premise	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Using fuels has negative impact on environment	.193	.003	-.647	.018	.060	.198	-.938	.001
CPL (WTU, SWEPCO) is doing enough for renewable energy and conservation	.279	<.001	.092	.382	.032	.311	.274	.195
Higher electric bills hinder economic development	-.888	<.001	.413	.094	.180	.008	-.532	.058
Higher electric bills lead people to use less electricity	.055	.209	.127	.339	.056	.224	.586	.056
Increasing use of renewable will make it less expensive in future	-.026	.377	-.017	.482	.654	<.001	-.716	.026
Likelihood ratio	-1254				-1930			

Note: All *p*'s are from one-tailed tests.

Table 9
Willingness to Pay More for Specific Options as a Function of Knowledge

A. Renewable Energy (CPL, WTU, SWEPCO)

Predictor	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Knowledge	-.102	.083	-1.103	.001	.207	<.001	-1.794	<.001
Likelihood ratio	-2120				-3189			

B. Conservation (CPL, WTU, SWEPCO)

Predictor	T1				T2			
	<u>Poisson model</u>		<u>Zero-inflation model</u>		<u>Poisson model</u>		<u>Zero-inflation model</u>	
	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
Knowledge	-.572	<.001	-1.087	.001	.284	<.001	-1.044	.002
Likelihood ratio	-2081				-2258			

Note: All *p*'s are from one-tailed tests.